



GIRRAWEEEN HIGH SCHOOL
EXAMINATION

2007

MATHEMATICS

Time allowed – 90 minutes

Year 11

HALF YEARLY

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Each question attempted is to be returned on a *separate* piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.

Question 1 (12 Marks)

Marks

- (a) Find $(1.7)^{23} \div 0.034$ expressing your answer in scientific notation correct to 4 significant digits. 2
- (b) Evaluate $\sqrt[3]{\frac{6.7+4.3}{9.6-1.5}}$ correct to 3 decimal places. 2
- (c) $\sqrt{1800} = x\sqrt{2}$ Find the value of x . 2
- (d) Write each of the following decimals as fractions in lowest terms
- (i) $0.\dot{6}\dot{3}$ 3
- (ii) $0.\dot{4}0\dot{4}$ 3

Question 2 (24 Marks)

- (a) Expand and simplify
- (i) $(\sqrt{3} + 2\sqrt{7})(\sqrt{14} - \sqrt{6})$ 3
- (ii) $(2\sqrt{7} - 3\sqrt{2})^2$ 3
- (b) Express with a rational denominator
- $\frac{3\sqrt{2} - 2\sqrt{3}}{5\sqrt{5} + \sqrt{7}}$ 3
- (c) Factorise fully
- (i) $8 - 18x^2$ 2
- (ii) $x^2 - 7x + 12$ 2
- (iii) $3x^2 - 10x + 3$ 3
- (iv) $8 - 27x^3$ 3
- (d) Simplify
- (i) $\frac{3a}{7} \times \frac{14}{2a^2}$ 2
- (ii) $\frac{x^2 - 6x + 8}{x - 2} \times \frac{x^2 - 9}{x^2 - 7x + 12}$ 3

Question 3 (22 Marks)

Marks

(a) Simplify

$$(i) \frac{3}{x^2-9} - \frac{2}{x-3} \quad 3$$

$$(ii) \frac{3x+3y}{x^2-y^2} \quad 2$$

(b) Solve the following

$$(i) \quad 3(2x-5) = 6-x \quad 2$$

$$(ii) \quad \frac{x-7}{2} \leq 3 + \frac{2x}{3} \quad 3$$

$$(iii) \quad |2x-3| = 5 \quad 3$$

$$(iv) \quad |3x-5| \geq 4 \quad 3$$

(c) Solve the simultaneous equations

$$(i) \quad \begin{aligned} 2a + 3b &= 4 \\ a - b &= 7 \end{aligned} \quad 3$$

$$(ii) \quad \begin{aligned} y &= x^2 - 3x \\ y + 2x &= 2 \end{aligned} \quad 3$$

Question 4 (12 Marks)

(a) Find correct to 3 decimal places

$$(i) \quad \tan 48^{\circ}07' \quad 1$$

$$(ii) \quad \sin 69^{\circ}31' \quad 1$$

(b) Find θ in degrees and minutes if $0 \leq \theta \leq 90^{\circ}$ and

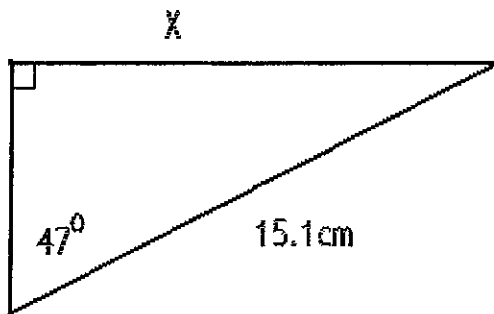
$$(i) \quad \cos \theta = \frac{\sqrt{5}}{4} \quad 2$$

$$(ii) \quad \sec \theta = \frac{3}{2} \quad 2$$

Marks

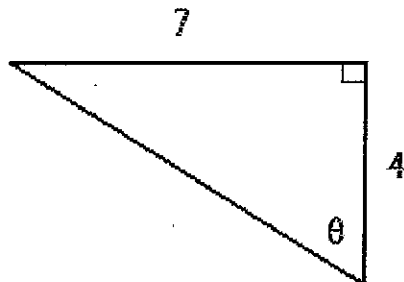
(c) (i) Find the value of x to one decimal place

3



(ii) Find θ to the nearest degree

3



Question 5 (16 Marks)

(a) Find the exact value of

- | | | |
|-------|-------------------------------|---|
| (i) | $\cos 30 + \sin 30$ | 1 |
| (ii) | $\tan^2 60$ | 1 |
| (iii) | $2 \sec 30$ | 1 |
| (iv) | $\tan 240$ | 1 |
| (v) | $\cos 135$ | 1 |
| (vi) | $2 \sin 30 \cos 30 - \sin 60$ | 2 |

(b) Solve the following equations for θ , $0 \leq \theta \leq 360^\circ$

- | | | |
|-------|---|---|
| (i) | $\sin^2 \theta = \frac{1}{4}$ | 3 |
| (ii) | $\tan^2 \theta = \frac{1}{3}$ | 3 |
| (iii) | $6 \cos^2 \theta + \cos \theta - 1 = 0$ | 3 |

Question 6 (18 Marks)

- (a) From the top of a tall building, the angle of depression to a point 60m from and on the same level as the base of the building is 82° . 3
Find the height of the building to the nearest metre.
- (b) A yacht sails from Sydney (S) on a bearing of 060° for 70km to a point L. It then sails on a bearing of 173° for 50km to H.
- (i) Draw a neat sketch of the course showing all the given information 3
 - (ii) Explain why the $\angle SLH$ is 67° 2
 - (iii) Using the Cosine Rule find the distance from H back to Sydney. 4
 - (iv) Using the Sine Rule find the angle $\angle LHS$ 4
 - (v) Hence or otherwise find the bearing back to Sydney from H. 2

Question 7 (9 Marks)

- (a) Prove the following identities
- (i) $\cos^4 \theta (\tan^2 \theta + 1) = 1 - \sin^2 \theta$ 2
 - (ii) $\frac{\sin \theta}{1 - \cos \theta} + \frac{\sin \theta}{1 + \cos \theta} = 2 \sec \theta \cot \theta$ 3
- (b) Draw a graph of the function $y = \cos x$. On the same axes graph the function $y = \sec x$, $0 \leq x \leq 180$ 4

Q1 (12 MARKS)

$$a) (1.7)^{23} \div 0.034 = 5872814.383$$

$$= 5.873 \times 10^6 \quad (2)$$

$$b) \sqrt[3]{\frac{6.7+4.3}{9.6-1.5}} = 1.10739 \quad \text{CALC}$$

$$= 1.107 \quad (2)$$

$$c) \sqrt{1800} = \sqrt{900 \times 2}$$

$$= 30\sqrt{2}$$

$$\therefore x = 30 \quad (2)$$

$$(i) x = 0.6\dot{3}$$

$$10x = 6.3\dot{3}$$

$$9x = 5.7$$

$$x = 5\frac{7}{90}$$

$$x = 19\frac{1}{30} \quad (3)$$

$$(ii) x = 0.40\dot{4}$$

$$1000x = 404.40\dot{4}$$

$$x = 0.40\dot{4}$$

$$999x = 404$$

$$x = \frac{404}{999} \quad (3)$$

Q2 (24 MARKS)

$$(a) (i) (\sqrt{3} + 2\sqrt{7})(\sqrt{14} - \sqrt{6})$$

$$= \sqrt{42} - 3\sqrt{2} + 14\sqrt{2} - 2\sqrt{42}$$

$$= 11\sqrt{2} - \sqrt{42} \quad (3)$$

$$(ii) (2\sqrt{7} - 3\sqrt{2})^2 = 28 - 12\sqrt{14} + 18$$

$$= 46 - 12\sqrt{14} \quad (3)$$

$$(b) \frac{3\sqrt{2} - 2\sqrt{3}}{5\sqrt{5} + \sqrt{7}} \times \frac{5\sqrt{5} - \sqrt{7}}{5\sqrt{5} - \sqrt{7}}$$

$$= \frac{15\sqrt{10} - 3\sqrt{14} - 10\sqrt{15} + 2\sqrt{21}}{(5\sqrt{5})^2 - (\sqrt{7})^2}$$

$$= \frac{15\sqrt{10} - 3\sqrt{14} - 10\sqrt{15} + 2\sqrt{21}}{118} \quad (3)$$

$$(c) (i) 8 - 18x^2 = 2(4 - 9x^2)$$

$$= 2(2 - 3x)(2 + 3x) \quad (2)$$

$$(ii) x^2 - 7x + 12 = (x - 3)(x - 4) \quad (2)$$

$$(iii) 3x^3 - 10x + 3 = (3x - 1)(x - 3) \quad (3)$$

$$(iv) 8 - 27x^3 = (2 - 3x)(4 + 6x + 9x^2) \quad (3)$$

$$(d) (i) \frac{3a}{7} \times \frac{14}{2a^2} = \frac{3}{a} \quad (2)$$

$$(ii) \frac{x^2 - 6x + 8}{x - 2} \times \frac{x^2 - 9}{x^2 - 7x + 12}$$

$$= \frac{(x - 4)(x - 2)(x - 3)(x + 3)}{(x - 2)(x - 3)(x - 4)}$$

$$= x + 3 \quad (3)$$

Question 3 22 MARKS

$$(i) \frac{3}{x^2-9} = \frac{2}{x-3}$$

$$= 3 - 2(x+3)$$

$$\frac{3}{x^2-9}$$

$$= \frac{-2x-3}{x^2-9}$$

(3)

$$(ii) \frac{3x+3y}{x^2-y^2} = \frac{3(x+y)}{(x-y)(x+y)}$$

$$= \frac{3}{x-y}$$

(2)

$$(i) 3(2x-5) = 6-x$$

$$6x-15 = 6-x$$

$$7x = 21$$

$$x = 3$$

(2)

$$(ii) \frac{x-7}{2} \leq 3 + \frac{2x}{3}$$

$$3(x-7) \leq 2(9+2x)$$

$$3x-21 \leq 18+4x$$

$$-39 \leq x$$

$$x \geq -39$$

(3)

$$iii) |2x-3| = 5$$

$$2x-3 = 5 \text{ OR } 2x-3 = -5$$

$$2x = 8$$

$$x = 4$$

$$2x = -2$$

$$x = -1$$

(3)

$$iv) |3x-5| \geq 4$$

$$3x-5 \geq 4 \text{ OR } 3x-5 \leq -4$$

$$3x \geq 9$$

$$x \geq 3$$

$$3x \leq 1$$

$$x \leq \frac{1}{3}$$

(3)

$$(c) (i) 2a+3b = 4 \quad (1)$$

$$a-b = 7 \quad (2)$$

$$\text{From (2) } a = 7+b$$

$$\text{Subst (1) } 2(7+b)+3b = 4$$

$$14+5b = 4$$

$$b = -2$$

$$a = 5$$

(3)

$$(ii) y = x^2-3x \quad (1)$$

$$y = -2x+2 \quad (2)$$

$$x^2-3x = -2x+2$$

$$x^2-x-2 = 0$$

$$(x-2)(x+1) = 0$$

$$x = -1, 2$$

$$y = 4, -2$$

$$(-1, 4) \quad (2, -2) \quad (3)$$

Question 4 (12 MARKS)

$$(i) \sin 48^\circ 07' = 1.115 \quad (1)$$

$$(ii) \sin 69^\circ 31' = 0.937 \quad (1)$$

$$(b) (i) \cos \theta = \frac{\sqrt{5}}{4} \quad \theta = 56^\circ 1' \quad (2)$$

$$(ii) \sec \theta = \frac{3}{2} \quad \cos \theta = \frac{2}{3}$$

$$\theta = 48^\circ 11' \quad (2)$$

$$(c) (i) \sin 47 = \frac{x}{15.1}$$

$$x = 15.1 \sin 47$$

$$x = 11.0 \text{ cm}$$

(3)

$$(ii) \tan \theta = \frac{7}{4}$$

$$\theta = 60^\circ$$

(3)

Question 5. 16 MARKS

2) (i) $\cos 30 + \sin 30$

$$= \frac{\sqrt{3}}{2} + \frac{1}{2} = \frac{\sqrt{3}+1}{2} \quad (1)$$

(ii) $\tan^2 60 = (\sqrt{3})^2$
 $= 3 \quad (1)$

(iii) $2 \sec 30 = \frac{2}{\cos 30}$
 $= \frac{2}{\frac{\sqrt{3}}{2}}$
 $= \frac{4}{\sqrt{3}} \quad (1)$

(iv) $\tan 240 = \tan(180+60)$
 $= \tan 60$
 $= \sqrt{3} \quad (1)$

(v) $\cos 135 = \cos(180-45)$
 $= -\cos 45$
 $= -\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2} \quad (1)$

(vi) $2 \sin 30 \cdot \cos 30 = \sin 60$
 $= 2 \times \frac{1}{2} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$
 $= 0 \quad (2)$

b) (i) $\sin^2 \theta = \frac{1}{4}$

$$\sin \theta = \pm \frac{1}{2}$$

$\theta = 30, 150, 210, 330 \quad (3)$

(ii) $\tan^2 \theta = \frac{1}{3}$
 $\tan \theta = \pm \frac{1}{\sqrt{3}}$

$\theta = 30, 150, 210, 330 \quad (3)$

(iii) $6 \cos^2 \theta + \cos \theta - 1 = 0$

$$(3 \cos \theta - 1)(2 \cos \theta + 1) = 0$$

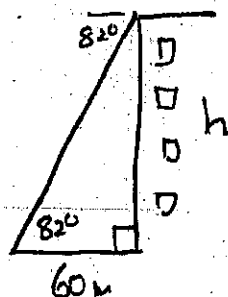
$$\cos \theta = \frac{1}{3} \text{ or } -\frac{1}{2}$$

$$\theta = 70.5^\circ, 289.5^\circ$$

$$\theta = 120^\circ, 240^\circ \quad (3)$$

Question 6. 18 MARKS

(a)



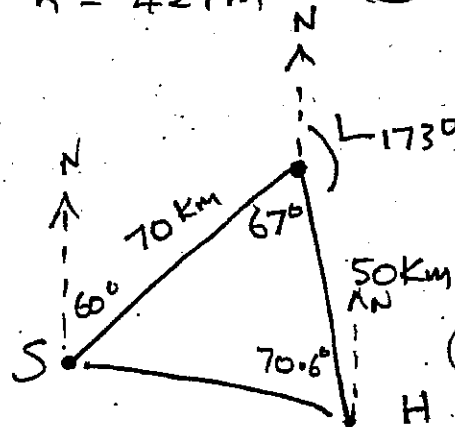
$$\tan 82 = \frac{h}{60}$$

$$h = 60 \tan 82$$

$$h = 426.9$$

$$h = 427 \text{ m} \quad (3)$$

(b) (i)



(ii) $\angle SLN = 120^\circ$ $\angle NLH = 173^\circ$

$$\therefore \angle SLH + 120^\circ + 173^\circ = 360^\circ$$

$$\angle SLH = 67^\circ \quad (2)$$

(iii) $l^2 = h^2 + s^2 - 2hs \cos L$

$$l^2 = 70^2 + 50^2 - 7000 \cos 67$$

$$l^2 = 4664.88$$

$$l = 68.3 \text{ km} \quad (4)$$

(iv) $\frac{\sin H}{h} = \frac{\sin L}{l}$

$$\frac{\sin H}{70} = \frac{\sin 67}{68.3}$$

Question 6 cont.

$$\sin H = 0.9434$$

$$H = 70.6^\circ$$

(4)

$$1) \angle LHN = 7^\circ \quad \angle LHS = 70.6^\circ$$

$\therefore$ Bearing H to Sydney

$$= 360 - 77.6$$

$$= 282.4^\circ$$

Question 7 (9 marks) (2)

$$1) (i) \cos^4 \theta (\tan^2 \theta + 1) = LHS$$

$$= \cos^2 \theta (\sin^2 \theta + \cos^2 \theta)$$

$$= \cos^2 \theta$$

$$= 1 - \sin^2 \theta = RHS \quad (2)$$

$$(ii) \frac{\sin \theta}{1 - \cos \theta} + \frac{\sin \theta}{1 + \cos \theta} = RHS$$

$$= \frac{\sin \theta (1 + \cos \theta) + \sin \theta (1 - \cos \theta)}{1 - \cos^2 \theta}$$

$$= \frac{\sin \theta + \sin \theta \cos \theta + \sin \theta - \sin \theta \cos \theta}{1 - \cos^2 \theta}$$

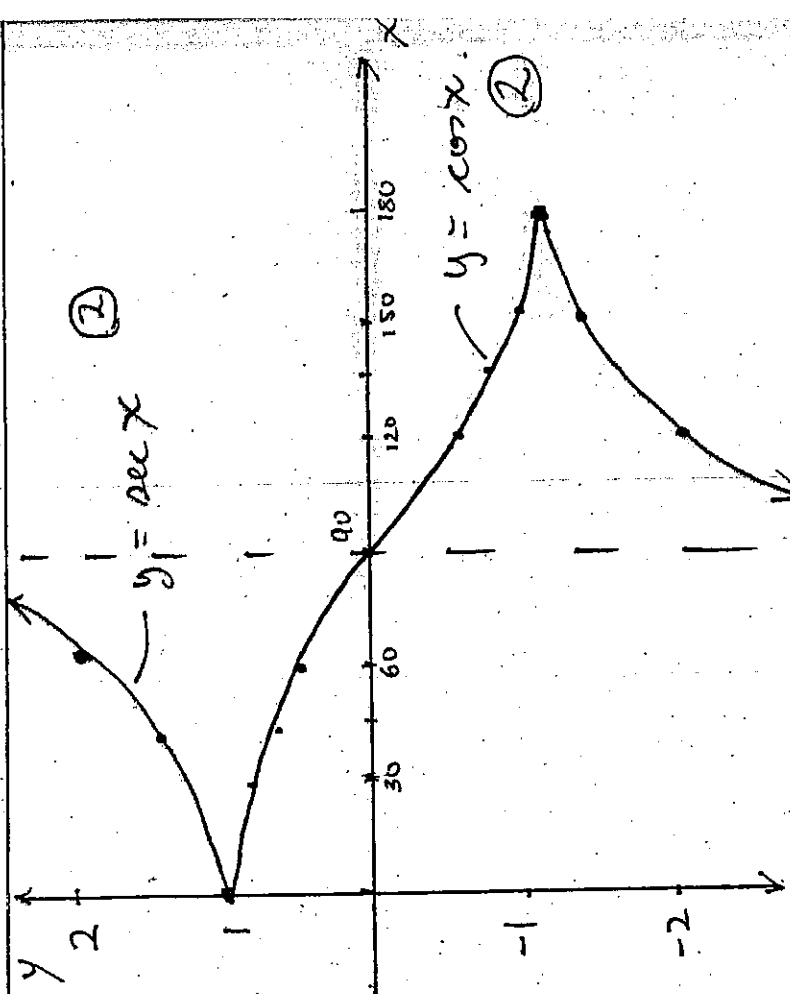
$$= \frac{2 \sin \theta}{1 - \cos^2 \theta}$$

$$= \frac{2}{\sin \theta}$$

$$= \frac{2 \cos \theta}{\sin \theta \cos \theta}$$

$$= 2 \cot \theta \sec \theta = RHS.$$

(3)



7(b)